Hybrid Manufacturing with Directed Energy Deposition 261

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Class Outline

Introduction to Directed Energy Deposition
Directed Energy Deposition Methods
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Production Challenges of Standard DED
Review: DED Introduction
Hybrid Directed Energy Deposition
DED Integration with CNC
Advantages of Hybrid DED
Hybrid DED Applications: Building Net-Shape Parts
Hybrid DED Applications: Adding Part Features
Hybrid DED Applications: Repairing Damaged Parts
Hybrid DED Applications: Applying Material Coatings
Final Review: Hybrid DED Applications

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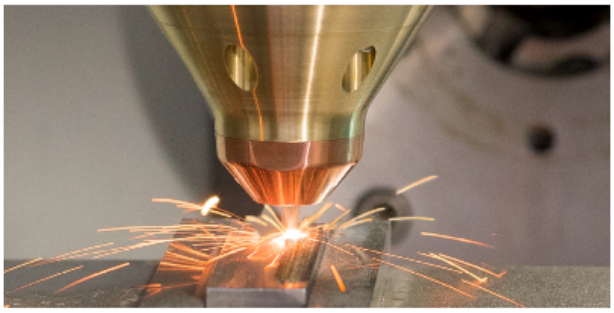
Introduction to Directed Energy Deposition



Directed energy deposition (DED) is an additive manufacturing (AM) process that builds parts using a thermal energy source to melt raw material from powder or wire feedstock onto a surface. Depending on the DED method, the raw material may be dispensed as feedstock through a separate device or through a deposition head that contains the heat source. Many DED systems use a robotic arm to control the tool movements. However, in a hybrid additive manufacturing (hybrid AM) application, a deposition head works as a print head attachment for a computer numerical control (CNC) machine or as a pre-installed component on an all-in-one hybrid machining center. Hybrid AM combines additive with traditional CNC machining methods on one hybrid machine. Due to its high deposition speed and relatively simple integration, DED is the most commonly used AM technology in hybrid AM.

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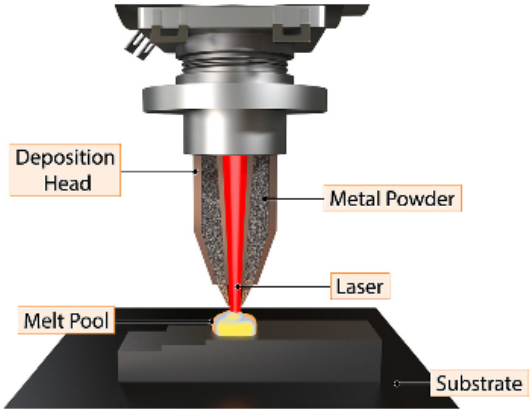
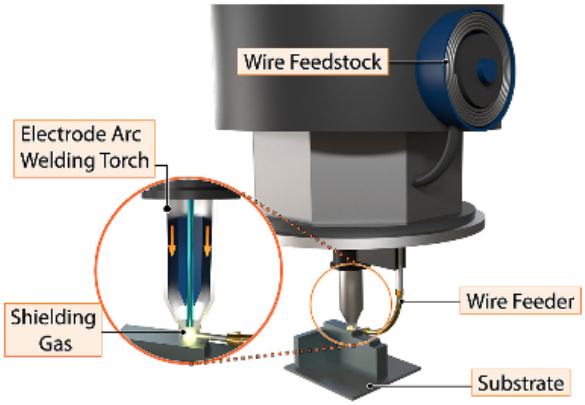
Directed Energy Deposition Methods



Laser Metal Deposition

The most commonly used DED method in hybrid AM machines is laser metal deposition (LMD). LMD, also known as direct metal deposition (DMD) or laser deposition welding (LDW), uses a laser as a thermal energy source. In LMD, a material hopper dispenses powdered metal through channels in a deposition head. A laser inside the head melts the powdered metal into a melt pool using intense heat. LMD typically uses a shielding gas to prevent contamination of the melt pool. Because LMD emits both heat and raw material from a single device and does not require a ground, it is the easiest to integrate and deploy inside a CNC machine.



Wire Arc Additive Manufacturing

An electric arc, as used in conventional welding, is also a common thermal energy source for DED. The most common welding tools for hybrid DED are gas metal arc welding (GMAW) tools, which use the feedstock as a consumable electrode to create the arc. Some hybrid machines use gas tungsten arc welding (GTAW) tools, which create the arc using a tungsten, nonconsumable electrode. Arc welding uses a shielding gas and typically uses wire as a feedstock. When used for DED applications, it is often promoted as wire arc additive manufacturing (WAAM). Since wire feedstock is usually a lower cost than metal powder, WAAM is often used to produce larger builds than powder DED methods. However, arc energy sources can also use metal powder as a feedstock.

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Production Benefits of DED

Typically, DED allows faster build rates than other metal AM methods such as powder bed fusion (PBF). DED tools also create larger builds than PBF, giving manufacturers a wider range of part size options. Often, DED can be used to deposit thin wall components more effectively than other manufacturing methods. As a result, DED can be ideal for building large, lightweight metal parts. Additionally, DED requires far less feedstock material to run than a PBF system, so upfront costs for materials are lower.

By controlling temperature and other parameters for DED, manufacturers can improve the microstructure of the metal as it cools to optimize mechanical properties like strength and hardness. DED tools can also deposit a variety of metals from different feedstocks in precise areas of a workpiece. This optimizes the production of parts containing multiple metal types and also enables deposition of metal matrix composites (MMCs). A manufacturer can model a new part containing a variety of metals and MMCs, and the DED tool can build the part directly from the model in a single process.

DED tools can also print additional material layers onto an existing workpiece effectively. This allows manufacturers to customize existing parts in order to optimize certain mechanical properties and improve performance. For example, a manufacturer may use corrosion-resistant metals like cobalt or nickel to build areas of a part that typically experience heavy stress or wear during use, such as a flange or bearing component.

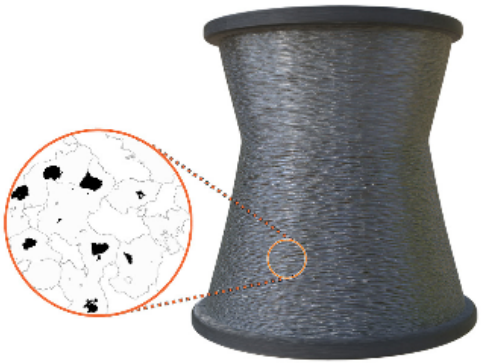

Figure 1: Adjusting DED operating parameters can potentially improve the microstructure of part builds.

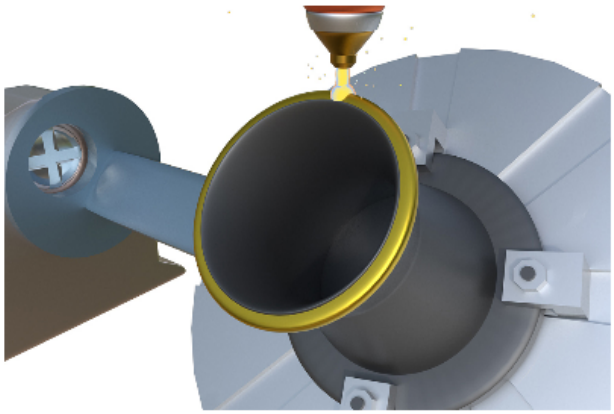

Figure 2: DED machine tools can apply multiple metals in a single build process.

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Production Challenges of Standard DED

The benefits of DED also come with a number of production challenges. Like other AM methods that use melting, the deposited metal shrinks as it cools. For this reason, part features that require precision such as holes, threads, and mating parts are difficult to produce without post-processing methods like machining. Metal shrinkage can also cause variability from one part to the next. However, building thin-walled, symmetrical parts with DED helps keep cooling uniform to reduce variability and also balance out thermal stresses that occur during the build process.

Unlike PBF systems where the build rests in a bed of reusable powder that can support various overhanging part features, standard DED systems cannot normally produce overhanging part features that are greater than a 45-degree angle without building support structures. However, while non-metal AM processes like fused deposition modeling (FDM) often use support structures, most DED applications do not use support structures because they create metal waste that can negate the cost savings of using DED. Manufacturers can use multi-axis machines to reorient the part in order to produce some overhangs. However, even on a multi-axis machine, DED still cannot create planar bridging features. This and the larger scale resolution of DED means that parts are typically less complex than those produced by PBF.

While DED tools have faster build rates than other AM methods, they also print part surfaces at a lower resolution. However, this and many other challenges of DED can be overcome with hybrid manufacturing.

Figure 1: Low print resolution results in a rough part surface.

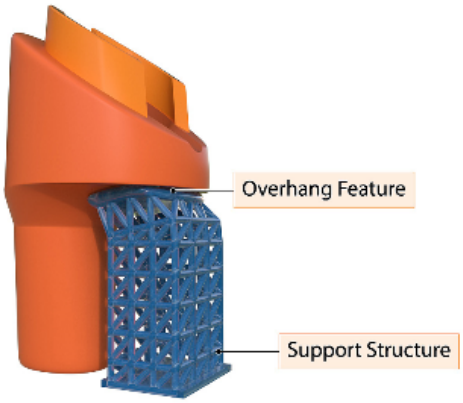

Figure 2: FDM parts commonly use support structures that would be costly with DED.

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Review: DED Introduction

1. PBF is the most commonly used additive manufacturing technology in hybrid AM. **FALSE**
2. Directed energy deposition (DED) typically has faster build rates than other additive manufacturing methods. **TRUE**
3. Directed energy deposition (DED) prints parts at higher resolution than most additive manufacturing methods. **FALSE**
4. Wire arc additive manufacturing, a type of directed energy deposition, requires a shielding gas. **TRUE**
5. Directed energy deposition machines use a powder bed to support building complex parts. **FALSE**
6. Laser metal deposition is the most widespread type of directed energy deposition to install on a computer numerical control machine. **TRUE**

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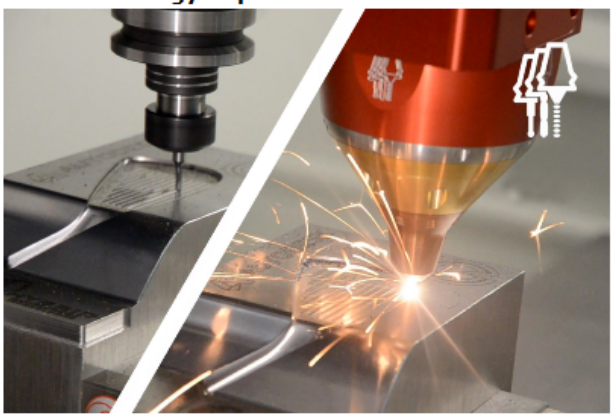
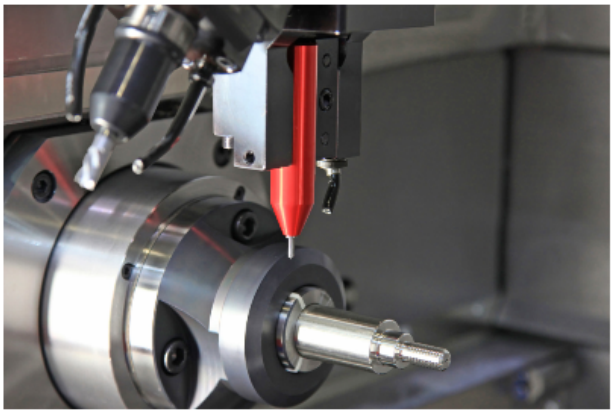
Hybrid Directed Energy Deposition



CNC Milling Setup, courtesy of Hybrid Manufacturing Technologies

In hybrid DED part production on a mill, the required deposition heads can be installed alongside the spindle or loaded directly into the automatic tool changer (ATC), which can then load it into the spindle for operation. Loading the head in the spindle allows for deposition directly on the spindle centerline. The ability to use multiple heads in the spindle also enables greater flexibility for different processing types. Before running a part program, an operator secures a substrate or workpiece onto the worktable using a vise, clamps, or other workholding device. Then, the operator runs the part program, which controls the toolpaths and tool changes required to produce the part. The deposition head prints different layers of the part, and cutting tools remove material from the printed layers based on the part program, which is often specialized for hybrid DED.



CNC Turning Setup

Some hybrid DED systems use a CNC lathe or turning center setup. In hybrid DED part production on a lathe, deposition heads are installed in a tool turret, often with a live tooling setup alongside cutting tools. An operator then secures a cylindrical workpiece in a three- or four-jaw chuck. The chuck is attached to a spindle which rotates the workpiece at different speeds based on commands in a part program or operator commands. The spindle rotates slowly or stops while the deposition head in the turret prints layers of material onto the workpiece, and it rotates at higher speeds while the turret moves stationary cutting tools into the workpiece. Layers of material are added and machined based on the part program.

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DED Integration with CNC

DED is one of the lowest resolution AM techniques available, which enables it to have one of the highest deposition rates of all AM techniques. However, these high deposition speeds also produce parts with the least fidelity to the digital model of the desired part. For this reason, parts made by DED virtually always require building slightly larger than the model dimensions, then machining as a post-processing step. This two-step approach can be simplified into the same setup on a hybrid DED and CNC machine.

Of the various DED energy sources and options available, LMD is currently the most widely adopted for use inside of hybrid machines. This is partly because using a laser as the energy source provides the highest quality metal deposition with the lowest heat input into the substrate, resulting in fine-grained microstructure. Using a laser also eliminates the need for grounding the part as is required for arc-based DED approaches. Since DED deposits metal at only a single point, the existing G code and M code programming style, which controls toolpaths, tool changes, and other machine functions, is well suited for programming DED. As a result, traditional CNC programmers and machine operators will require less technical training when converting from traditional machining to a hybrid LMD process.

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Advantages of Hybrid DED

The faster build rates of traditional DED tools come at the expense of lower print resolutions. However, hybrid DED tools take advantage of the faster build rates while using subtractive machining methods to compensate for low resolution. Operators can simply use DED to build different areas of a part slightly larger than specifications, then machine each area to the specified dimensions. This also addresses variability effectively, as machining each part down to size cuts away any size variations that may occur during cooling. Hybrid DED thus allows manufacturers to build parts with tight tolerances faster and more efficiently than other methods.

DED typically minimizes the use of support structures since it creates metal waste. Since there is also no powder bed or built-in support method, this can limit the types of part features traditional DED tools can produce. However, hybrid DED tools that use additional axes to rotate parts can compensate for this drawback. Four-axis or five-axis machining permits the movement of tools in three axes as well as the rotation of the workpiece. With four or five-axis flexibility, deposition heads can access parts from multiple angles, so operators can print on layers of the part that are unreachable otherwise.

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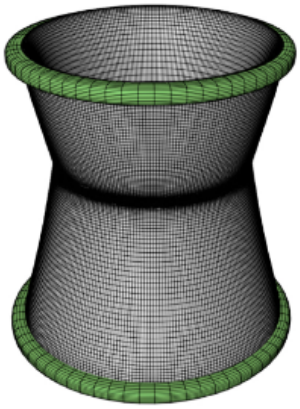
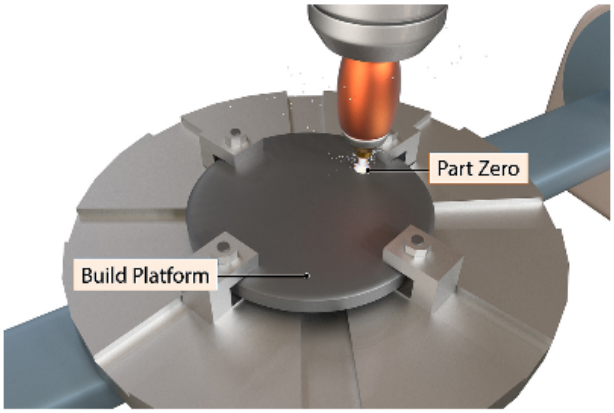
Hybrid DED Applications: Building Net-Shape Parts



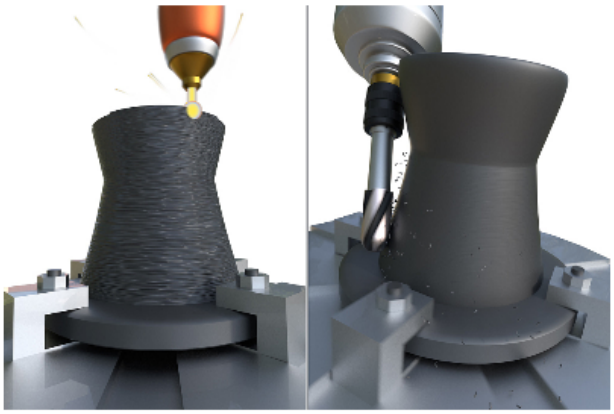
CAD software is used to model AM parts and create STL files.

To create a net-shape part with hybrid DED, a manufacturer first designs a specialized part, such as a rocket nozzle, using computer-aided design (CAD) software to create an stereolithography (STL) file. The manufacturer then uses a computer-aided manufacturing (CAM) software package that can translate the STL file into a part program that includes toolpaths and tool changes for the deposition head as well as machining and finishing tools. Hybrid DED typically uses G code part programs.



Part zero determines the starting point for a build on a hybrid DED and CNC machine.

Next, an operator or robot loads the appropriate tools and material feedstock onto the machine. The operator then establishes part zero on the subtrate or build platform, which is the bottom of the part and acts as the origin point for all machine movements in the part program.



DED and machining tools alternate to complete the part.

Finally, the deposition head deposits layers of material onto the substrate or build platform, and cutting tools machine the layers to correct dimensions based on the CAD model. The part program often alternates between DED and machining until the part is built and finished to net shape.

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Hybrid DED Applications: Adding Part Features

While hybrid DED can build specialized, net-shape parts effectively, it is more commonly used to build part features onto an existing workpiece. In this process, a manufacturer creates a CAD model of a part, such as a jet engine turbine. Next, hybrid CAM software converts the 3D model of the desired part feature, the turbine blades, into programmed toolpaths for the deposition head and finishing tools. The workpiece, an unpopulated rotor, is then loaded onto the hybrid machine using a workholding device. After setting part zero on the workpiece, the operator runs the program. Using precise machine coordinates in the part program, the deposition head prints the part feature onto the workpiece, and cutting tools apply finishing operations as needed to complete the part feature.

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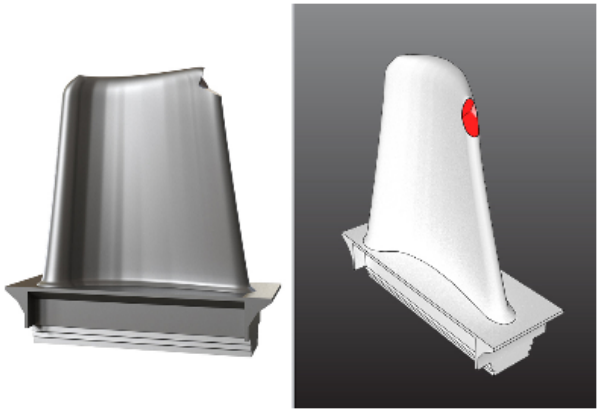
Hybrid DED Applications: Repairing Damaged Parts



Repairing metal parts is the most common use of DED. While welding and manual grinding is the traditional method for metal repair, hybrid DED allows manufacturers to automate traditional repair processes, such as repairing a damaged or worn turbine blade, blisk, or impeller and finishing the repaired surface on the same machine. To repair a single part, a technician may scan the damaged part with a laser scanner or touch probe to create a digital model, which is used as the model for the repair. Before repairing the part, a machining tool must remove the worn or damaged material to produce a clean surface at a known dimension. This machined surface becomes the initial build surface for adding material to rebuild the part using DED. Since DED produces a relatively rough surface finish, the deposition process should always overbuild the replaced part so that it is slightly larger than the final part dimensions, or near net shape. Lastly, a finishing tool removes a precise amount of material to finish the printed part surfaces to the part's final size, or net shape, using subtractive methods. A critical aspect of final machining is that the added repair surfaces should blend into the original part surfaces with very little mismatch. In a hybrid DED and CNC machine, this mismatch can often be minimized to less than 20 microns.

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Hybrid DED Applications: Applying Material Coatings

Like repair processes, manufacturers often use hybrid DED systems to add material coatings to parts and components. This process is often referred to as laser cladding. A material coating may be included as part of an original part design to increase wear resistance or other properties. Coatings may also be added to existing parts to add properties that were not present in the original part material, or that were lost due to wear. For example, coating a part with a superalloy can improve its strength, heat resistance, and other properties. Since superalloys are very expensive materials, coating the part costs much less than building the entire part from the superalloy. This is just one aspect of the overall cost savings, as applying certain coatings to part surfaces can also greatly extend the life of the part.

Unlike traditional cladding methods that use cold spraying or thermal spraying techniques, laser cladding with DED can create a metallurgical bond between the part and the coating metal. This metal bond is more durable than traditional cladding that only forms a mechanical bond. Laser cladding can also bond some dissimilar metals such as Inconel and steel more reliably and easily than alternative cladding methods. Laser cladding on a hybrid DED system also allows each layer to be machined to precise dimensions and finished to reduce surface asperities and variability before bonding the next type of metal.

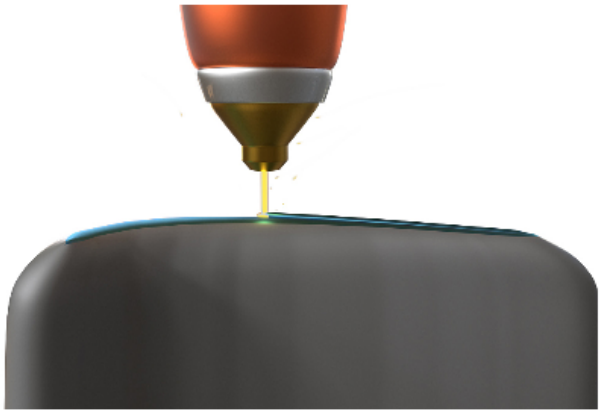

Figure 1: Laser Cladding

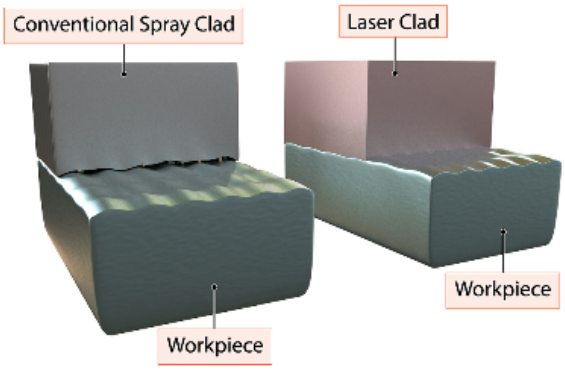

Figure 2:

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Final Review: Hybrid DED Applications
Which of the following refers to using DED tools to apply material coatings to a part?

- **Laser cladding**
- Part molding
- Net shaping
- Lightweighting

Which of the following is required to add features to an existing part using hybrid DED?

- **Workholding device**
- Part mold
- Substrate
- Build platform

Which application requires a hybrid DED process instead of a traditional DED process?

- **Building net-shape parts**
- Adding part features
- Building complex geometry
- Adding material coatings

What hybrid DED process requires technicians to scan an existing part to create a digital model rather than using CAD software?

- **Repairing part components**
- Building net-shape parts
- Adding part features
- Improving part tolerances

What disadvantage of standard DED can be overcome using hybrid DED?

- **Low print resolution**
- Slow deposition speeds
- Excess material waste
- High operating temperatures

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Class Vocabulary

Additive Manufacturing	AM. The process of joining or solidifying materials to make an object from a three-dimensional computer model. Additive manufacturing methods typically build up layers of material to create an object.
AM	Additive manufacturing. The process of joining or solidifying materials to make an object from a three-dimensional computer model. AM methods typically build up layers of material to create an object.
Asperities	Microscopic peaks found on all surfaces. Contact between asperities on different surfaces causes friction.
Automate	Using mechanical systems that operate with limited human interaction. Automated systems include CNC machines and robotics.
Automatic Tool Changer	A CNC mill or machining center component that holds and changes tools during operation. Automatic tool changers reduce cycle times by automatically changing tools between functions.
Bearing	A friction-reducing device that allows one moving part to glide past another moving part without excess wear or friction.
Blisk	A component made of a rotor disk and multiple attached blades. Blisks, or bladed disks, are commonly used in aerospace parts.
Build Platform	The flat surface on which a part is additively manufactured. The build platform can either be a permanent machine surface from which parts are removed or a surface that can be removed from a machine once the build is complete.
Build Rates	The time it takes to build a part or the number of parts that can be made in a set period of time. Build rates, or production rates, for additive manufacturing (AM) are considerably slower than traditional manufacturing.
CAD	Computer-aided design. A method of designing two- and three-dimensional objects using computers and software. CAD is most often used to create part models for production.
CAM	Computer-aided manufacturing. The use of computer software that facilitates the development of part programs to produce a part. CAM software applications create an image of the workpiece and develop the program code from information that the programmer inputs.
Chuck	A device that holds a workpiece in place as it rotates on a lathe or other machine. The chuck commonly has three or four jaws that can be adjusted to fit various workpieces.
Cladding	Adding a protective layer of material to a previously manufactured workpiece to improve corrosion resistance or other beneficial properties. Cladding is traditionally done using welding tools that create a mechanical bond between two similar metals.
Clamps	A device that holds an object by applying downward or inward pressure. Clamps are often used as workholding devices in manufacturing.
CNC	Computer numerical control. A programmable system of software and hardware that directs the operation of a machine. CNC systems use mathematical data to direct machine movements.
Coatings	The application of surface treatments to a material. Coating processes change a material's mechanical properties.
Cobalt	A hard, gray metal that is highly magnetic. Cobalt is often added to manufactured parts or components because of its superior coating and adhesion characteristics.
Cold Spraying	A process in which powdered materials are sprayed, below their melting point and at very high speeds, onto a surface to protect it from wear, heat, and corrosion. Cold spraying deforms and mechanically bonds the sprayed materials to the intended surface.
Computer Numerical Control	CNC. A programmable system of software and hardware that directs the operation of a machine. Computer numerical control systems use mathematical data to direct machine movements.

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Computer-Aided Design	CAD. A method of designing two- and three-dimensional objects using computers and software. Computer-aided design is most often used to create part models for production.
Computer-Aided Manufacturing	CAM. The use of computer software that facilitates the development of part programs to produce a part. Computer-aided manufacturing software applications create an image of the workpiece and develop the program code from information that the programmer inputs.
Consumable Electrode	A device that conducts electricity from the contact tip to the arc and melts into the weld as a filler metal. Some consumable electrodes may also provide shielding that protects the arc and weld puddle.
DED	Directed energy deposition. An additive manufacturing process in which focused thermal energy is used to melt materials as they are deposited. DED is often used with powdered or wire metal feedstock.
Deposition Head	A nozzle that melts and dispenses material. Deposition heads are used in directed energy deposition (DED) processes.
Directed Energy Deposition	DED. An additive manufacturing process in which focused thermal energy is used to melt materials as they are deposited. Directed energy deposition is often used with powdered or wire metal feedstock.
Electric Arc	An area in which electricity jumps from an electrode on a tool to a workpiece or build platform. An electric arc melts the wire feedstock in wire arc additive manufacturing (WAAM).
Features	A defining characteristic on a component or part. Features include corners, edges, holes, and grooves.
Feedstock	Raw material that is used to make manufactured parts. Feedstock for additive manufacturing methods includes filaments, pellets, wire, and powder.
Five-Axis	Machine control that can execute movement along three linear axes and two rotational axes. Five-axis machining capabilities can allow hybrid DED deposition heads to access areas of a part that are difficult to reach otherwise.
Flange	A ring or collar surrounding a cylindrical component such as a shaft or toolholder. A flange is mounted to other components to provide support and rigidity.
Four-Axis	Machine control that can execute movement along four axes concurrently. Four axis machines can move machining tools along three linear axes while turning the workpiece around an additional rotational axis.
Fused Deposition Modeling	FDM. An additive manufacturing process that builds objects by extruding layers of material. FDM usually uses thermoplastic materials.
G Code	A programming language that pairs address letters with numerical values to form commands. G code is used to direct the path of tools on traditional and hybrid CNC machines.
Gas Metal Arc Welding	GMAW. An arc welding process in which the bare wire electrode and inert shielding gas are fed to the weld through a welding gun. Gas metal arc welding tools can also be used in additive manufacturing processes.
Gas Tungsten Arc Welding	GTAW. A very precise arc welding process that uses a non-consumable tungsten electrode. GTAW is also referred to as tungsten inert gas (TIG) welding.
Ground	A source that absorbs stray electrical charge. Ground may refer to the earth or another large conducting body, such as a motor housing, that serves as a source of zero electrical potential.
Hybrid Additive Manufacturing	Hybrid AM. A process that uses both traditional and additive manufacturing (AM) to create a finished part. Hybrid additive manufacturing involves combining additive and subtractive processes on a single machine.
Hybrid Machining Center	A type of CNC machine that is capable of both additive manufacturing and traditional subtractive machining. Hybrid machining centers typically resemble the basic structure of a traditional CNC mill.
Impeller	A rotor inside a tube or conduit used to increase pressure and flow of a fluid. Impellers are sometimes referred to as blowers.
Inconel	A superalloy made of nickel and chromium that is designed to perform well in extreme environments. Inconel is very difficult to machine and resists both oxidation and corrosion.
Laser Cladding	The process of fusing a layer of material onto a part surface using a laser. Laser cladding fuses together metallurgically, creating a more durable bond than traditional cladding methods.
Laser Metal Deposition	LMD. A directed energy deposition process that uses a laser beam to melt powdered metal as it is deposited on a build platform. Laser metal deposition is sometimes referred to as laser powder

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deposited on a build platform. Laser metal deposition is sometimes referred to as laser powder forming.

Laser Scanner The process of using a concentrated beam of light to generate data about the geometric shape of an object. Laser scanning is used to create a point cloud that can be turned into a 3D computer-aided design (CAD) model.

Lathe A machine used to create parts, often cylindrical, by removing material from a workpiece. A lathe holds a rotating workpiece while the cutting tool is either fed into the workpiece or along its side.

Live Tooling A feature that equips the tool posts or turret on a lathe with power-driven tools, such as end mills and drills. Live tools can perform machining operations off the part centerline while the workpiece is held in the spindle.

LMD Laser metal deposition. A directed energy deposition process that uses a laser beam to melt powdered metal as it is deposited on a build platform. LMD is sometimes referred to as laser powder forming.

M Code Lines of code in the G code programming language. M codes are used to program machine functions, such as coolant or tool changes, and are often machine- or manufacturer-specific.

Material Hopper A container used to hold raw materials for use in an industrial process. Material hoppers used in metal AM processes hold powdered metal.

Mating Parts An object that a workpiece has been manufactured to fit. Nuts and bolts are mating parts.

Mechanical Bond A joining of two material using an adhesive in which the adhesive material physically locks into the crevices of the surface. A mechanical bond is less sturdy than a metallurgical bond.

Mechanical Properties A characteristic that describes how a material reacts when subjected to a force that attempts to stretch, compress, bend, dent, scratch, or break it. Mechanical properties include strength, toughness, ductility, and hardness.

Melt Pool An area of liquefied material, usually metal. Melt pools are created in directed energy deposition (DED) in order to fuse powdered metal or metal wire in layers to form a solid part.

Metal Matrix Composites MMCs. A composite made from a metallic matrix and high-performance reinforcements. MMCs have high levels of strength, stiffness, and relatively high resistance to heat.

Metallurgical Bond A joining of two separate metals in a way that combines their physical and mechanical properties. Metallurgical bonds are stronger than mechanical bonds.

Microns A metric unit of measurement equal to one thousandth of a millimeter (0.000039 inches). Microns, also known as micrometers, are used to measure layer thickness for DED processes.

Microstructure The relationship between the small, individual crystals in a metal. Microstructure, also known as grain structure, changes when the metal is exposed to high temperatures.

Mill A rotating, multi-point cutting tool that is guided along a workpiece to create flat surfaces, slots, or other features. The term mill also refers to the machine that uses mills to perform milling operations on a workpiece.

Near Net Shape A part with dimensions that are close to its final specified form after initial manufacturing steps. Near-net-shape parts require little subsequent machining.

Net Shape A part that meets consumer or manufacturer specifications, including tolerance and surface finish, directly after it is manufactured. Net-shape parts do not need any further post processing after they are created.

Nickel A corrosion-resistant nonferrous metal. Nickel is commonly added to electrodes and stainless steels to prevent corrosion.

Nonconsumable Electrode An electrode which is not used up during the process of welding. Nonconsumable electrodes are used in GTAW are made from tungsten.

Part Program A series of instructions produced by CAM software and used by a CNC machine to perform the necessary sequence of operations to complete a part based on a CAD design. Part programs are typically written in G code.

Part Zero The starting point of X, Y, and Z coordinates for a specific workpiece. Part zero acts as a reference point for all part dimensions.

Post-Processing A finishing process used to clean, grind, or otherwise prepare a manufactured additive manufacturing

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part for shipping to a manufacturer or other consumer. Common additive manufacturing post-processing steps include removing support structures, improving surface finish, and bringing the part into tolerance.

Powder Bed Fusion PBF. An additive manufacturing method that uses thermal energy to melt together layers of powdered polymer, metal, ceramic, or other material. Powder bed fusion processes often use either lasers or electron beams as thermal energy sources.

Print Head Attachment A portable additive manufacturing component that holds and distributes part build material. Print head attachments can be installed on many traditional CNC machines to convert them into hybrid machines.

Resolution The amount of detail that an AM machine can build in a set amount of space. A higher resolution means that a machine can build more detailed parts.

Robotic Arm A programmable or remote-controlled mechanical device that simulates the movement of a human arm. Robotic arms are used in a variety of assembly and manufacturing applications.

Shielding Gas A gas that surrounds and protects melted metal from oxidation. Shielding gases are used in some directed energy systems.

Spindle The rotating component of the machine tool. In a milling operation, the spindle holds and rotates the toolholder and cutting tool.

Stereolithography STL. Computer files that represent 3D models as a series of interconnected triangles. Stereolithography files are converted from CAD files and are used in additive manufacturing machines.

Substrate A surface or medium that serves as a base for other materials or components. Some directed energy deposition processes deposit materials onto substrates rather than a build platform.

Superalloy An expensive, complex metal alloy designed to perform under intense conditions, such as elevated temperatures. Superalloys are also known as high performance alloys.

Support Structures A reinforcing component used to hold the weight of an additively manufactured part as it is being constructed. Support structures are removed from the part once the build is complete.

Thermal Energy Source A device that generates power in the form of heat. Thermal energy sources, such as devices that generate electron beams, lasers, and plasma arcs, are used during some additive manufacturing methods.

Thermal Spraying A process in which heated materials are sprayed, at or above their melting point, onto a surface to protect it from wear, heat, and corrosion. Thermal spraying can mechanically bond metal coatings to an existing surface.

Thermal Stresses Damage to a material due to excessive changes in temperature. Thermal stresses may occur during additive manufacturing methods, such as powder bed fusion (PBF) and directed energy deposition (DED), causing a part to warp or distort.

Tolerances An acceptable deviation from a desired dimension that meets specifications. Parts outside of a required tolerance must be discarded or reworked.

Tool Turret A lathe component that holds a number of cutting tools. The tool turret rotates to place tools in the cutting position.

Toolpaths A series of coordinate positions that determine the movement of a tool during a machining operation. Toolpaths are expressed in G code.

Toolpaths A series of coordinate positions that determine the movement of a tool during a machining operation. Toolpaths are expressed using G code commands.

Touch Probe A type of contact probe that detects a feature on a part and generates an electronic signal to record its dimension. Touch probes are commonly used to create 3D models for DED repair.

Turbine A machine for producing continuous power. A turbine consists of a wheel or rotor made to revolve by a fast-moving flow of water, steam, gas, air, or other fluid.

Turning Center A sophisticated CNC lathe that can perform a variety of drilling and milling operations all at the same location. Turning centers usually use live tooling.

Variability Inconsistencies in dimensions from one part to the next when producing the same part multiple times. Variability is a challenge that manufacturers must cope with when producing parts using additive manufacturing methods.

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Vise A workholding device with two jaws that grip and hold a workpiece in place. Vises are mostly used to hold rectangular workpieces with fairly simple dimensions.

Wear Resistance A material's ability to resist the gradual wearing away caused by abrasion and friction. Increased wear resistance can lengthen the life of a material.

Wire Arc Additive Manufacturing WAAM. A directed energy deposition (DED) process that uses an electric arc and a shielding gas to melt materials as they are deposited onto a build platform. Wire arc additive manufacturing typically uses wire feedstock.

Workholding A method or device for securing a workpiece for a machining operation. Workholding can include chucks, vises, and bolts.

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